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Key Points:

- An alternative configuration for the differential magnetometer measurement (DMM) method for monitoring geomagnetically induced currents (GICs) is presented
- This augmented DMM technique provides two independent component-wise measurements of GICs
- The inferred GIC is compared and validated against transformer neutral-to-ground current data from Alberta's electrical power grid

Supporting Information:

Supporting Information may be found in the online version of this article.

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Using a Differential Magnetometer Technique to Measure Geomagnetically Induced Currents: An Augmented Approach

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Abstract Geoelectric fields produced by time-varying magnetic fields during geomagnetic storms can result in potentially damaging geomagnetically induced currents (GICs) in long conductors at the Earth's surface. GICs can pose a significant risk to the integrity of grounded electrical infrastructure, particularly high-voltage transformers. In this study, an inferred GIC is calculated using an augmented differential magnetometer measurement (DMM) technique on a 500 kV transmission line in central Alberta and is validated using a proximal transformer neutral-to-ground (TNG) current measurement by AltaLink L.P. using a Hall probe at a transformer substation. This research outlines a custom-built and innovative DMM design by which both DMM sensors deployed around a power line measure the background geomagnetic disturbance (GMD) field and the magnetic field generated locally by the GIC. We show how this modified approach provides two independent estimates for GIC derived using only ΔB_y or ΔB_z, the magnetic field components perpendicular to the line carrying GIC. Results for a geomagnetic storm on 12 Oct 2021 show contemporaneous peaks in the TNG current and the DMM-inferred GIC. The two data sets have similar waveforms and are within the same order of magnitude. The background GMD is reconstructed using DMM and shows excellent correlation to the measured GMD at the permanent Canadian Array for Real-time Investigations of Magnetic Activity magnetic station at Ministik Lake, approximately 48.5 km away. Based on the results presented here, we verify the added utility value of DMM for temporary deployments for assessing GIC risk in electrical power grids.

Plain Language Summary During geomagnetic storms, changes in the Earth's magnetic field can result in geomagnetically induced currents (GICs) that travel through conductive structures at the Earth's surface, such as power lines. GICs are near-DC currents that can be harmful to electrical infrastructure. In this study, we built and tested an augmented differential magnetometer measurement (DMM) setup where sensors placed around the high-voltage power line could measure the natural magnetic field disturbances and the magnetic field fluctuations caused by GICs. Using the differential measurement in the components of the magnetic field perpendicular to the power line, two estimates for GIC were calculated during the geomagnetic storm on 12 October 2021, on a 500 kV transmission line in Alberta, Canada, and then compared to the Hall probe GIC data from a nearby AltaLink transformer substation. The results show that the inferred GICs from the DMM matched well with the Hall probe current measurements in waveform and amplitude. Also, the background magnetic disturbance measured by the DMM closely matched the background magnetic field measured at a permanent magnetic station about 48.5 km away. Based on these findings, we conclude that DMM could be useful for temporary deployments to assess GIC risk in electrical power grids.

1. Introduction

During space weather events, the Earth's magnetic field can become exhibit rapid time variations resulting in geomagnetic disturbances (GMD), a spatially and temporally varying magnetic field at the Earth's surface. Both the Earth and conductive infrastructure networks, such as power lines, pipelines and railroads, experience an induced electromotive force (emf) due to time variations in the geomagnetic field. This induced emf drives low frequency, $f < 1$ Hz, electrical currents called geomagnetically induced currents (GICs) (e.g., Boteler & Pirjola, 2017). GICs are excess quasi-direct currents and can flow through power transmission lines (e.g., Boteler et al., 1998; Pirjola, 2000), and pose a significant threat to the integrity of the electrical infrastructure. The introduction of a near-DC current in the electrical system can produce an offset of the magnetic flux density in the transformer core resulting in half-cycle saturation (Price, 2002). Saturation of transformer cores can lead to transformer heating, increased reactive power demand, incorrect protection relay operation, and can cause, in the most extreme cases, electrical blackouts (Molinski, 2002; Pirjola, 2000), such as the well-known Hydro-Quebec blackout in 1989 (Bolduc, 2002; Boteler, 2019). GICs have magnitudes that can range on the order of 1–100's of Amperes with the largest GIC recorded in Sweden during a magnetic storm on 2000-04-06 reaching almost 300 A (Boteler & Pirjola, 2017; Pulkkinen et al., 2003). Geophysical factors that influence GIC magnitudes are ground conductivity and the size of the GMD which statistically varies with latitude (Pirjola, 2000). There is also an increased vulnerability of power networks when subjected to larger and more complex GMD fields (Molinski, 2002). It is therefore important to estimate and measure GICs flowing within the electrical power network in order to better understand their effects on network reliability.

Direct measurements of GICs on electric power grid infrastructure can be made using a Hall effect sensor installed in the electrical substation at the transformer neutral-to-ground (TNG) connection. These sensors measure the DC component of the current flowing through the ground wire such that under geomagnetically quiet conditions the current flowing through the Earthing point is near zero. The use of Hall probes to monitor GICs in high voltage (HV) networks is still somewhat scarce. With the exception of New Zealand (MacManus et al., 2017; Marshall et al., 2012), data from Hall sensor measurements have only been reported from limited locations and in only a few electricity networks: for example, Japan (Watari et al., 2009), Ireland (Blake et al., 2018), Spain (Torta et al., 2017), Finland (Viljanen & Pirjola, 1994), Scotland (Thomson et al., 2005), Austria (Bailey et al., 2018), Sweden (Rosenqvist & Hall, 2019), and the USA (Butala et al., 2017). These studies primarily used GIC measurements for the validation of network response models.

An alternative approach to measure GICs in an element of the power grid is by using the differential magnetometer measurement (DMM) method. This method was first introduced by Mäkinen (1993) and further described by Viljanen and Pirjola (1994), field tested at low latitudes by Trivedi et al. (2007) in Brazil, Marsal et al. (2021) in Spain, and Matandirotya et al. (2016) on two 3-phase HV lines in Namibia, and expanded to more complex HV systems at mid-latitude in the UK by Hübert et al. (2020, 2024). Previous DMM applications have also included measuring GIC on pipelines (Campbell, 1980; Pulkkinen et al., 2001). The standard DMM configuration involves two fluxgate magnetometers, one placed directly underneath the power line referred to as the underline sensor, and the other placed at some distance perpendicular to the power line referred to as the remote sensor. Due to its specific placement, the underline sensor measures both the background magnetic field and the magnetic field caused by the GIC, while in the traditional configuration, the remote sensor is placed far enough away as to not record the magnetic field produced by the GIC, and therefore can serve as a reference station to which data from the underline sensor can be compared. The difference between the measurements from the two sensors is the magnetic field associated with the GIC, $\delta\mathbf{B}(t)$, which can be used to calculate the current flowing through the power line through an application of Ampère's law. The DMM method has been shown to be an economical alternative to TNG Hall sensor methods for measuring GIC (e.g., Hübert et al., 2020). Drawbacks of the Hall effect GIC measurements include accuracy, specialized power engineer time, the permanent installation of (typically) a limited number of sensor installations across a large network, the invasive nature of accessing the substation to perform the installations, temperature sensitivity, and sensitivity to DC offset from other events, and of which is often exacerbated by restrictions of access to this industry data. Many of these drawbacks are addressed by using a DMM method. This is a strong motivation to design, test and develop a DMM system capable of consistent and validated GIC measurements which can be non-invasively and relatively easily deployed at low cost. Since the DMM infrastructure is relatively easy to move, it can also be deployed sequentially at many locations in a network.

Here we focus on measurements of GIC arising from an application of an augmented DMM technique in Alberta, Canada. This has the advantage that a number of GMD monitoring stations are available from the Canadian Array for Real-time Investigations of Magnetic Activity (CARISMA; Mann et al., 2008), in addition to data from sensors in the DMM deployment. Due to the lack of data availability, particularly the general absence of transformer neutral-to-ground (TNG) current measurements by industry, only limited GIC modeling studies have previously been completed in Alberta using 1D (Haque et al., 2017; Nikitina et al., 2016) and 3D (Cordell et al., 2021) electric field models, with no validation against measured GIC. However, the recent installation of Hall effect sensors by AltaLink, Alberta's largest electric transmission company, provides an excellent opportunity to validate and compare the industry Hall effect GIC measurement and the GIC results in elements of the transmission network derived using the DMM method. AltaLink installed Hall effect sensors at the neutral-to-ground connection at transformer substations at six locations across the province, and began measurements

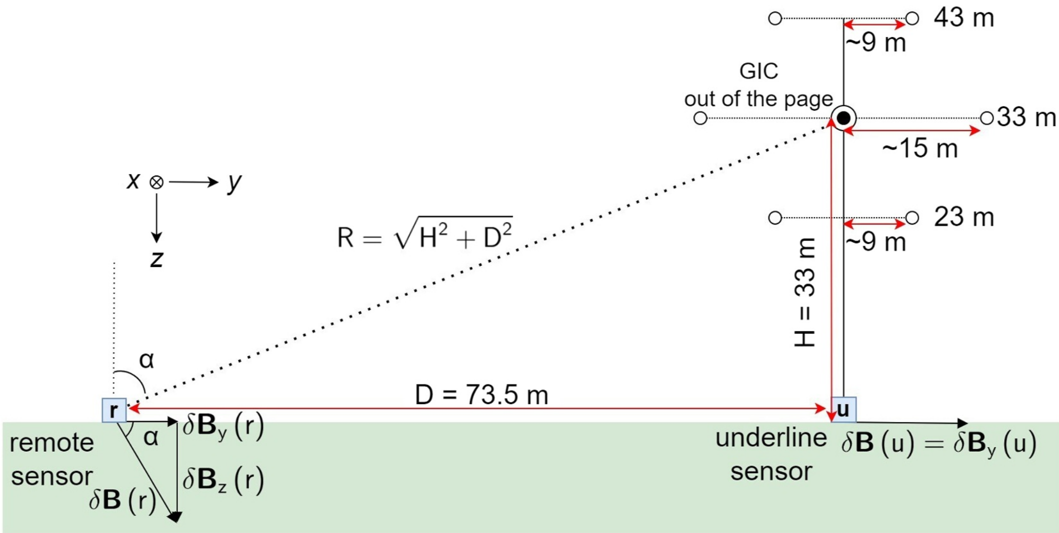

Figure 1. A schematic of the DMM deployment shows the relative location of each 3-axis fluxgate magnetometer, underline (u) and remote (r), with respect to the overhead power lines. Each magnetometer measures both the large scale background magnetic field of the GMD, $\mathbf{B}_0(t)$, and the magnetic field from GIC in the power lines, $\delta\mathbf{B}(t)$, in the x (Northward), y (Eastward) and z (down) components. The heights of the wires in the tower are assumed to be 23 m, 33 m, and 43 m. The span between pairs of wires is 18 m for the top and bottom rows and 30 m for the center row of wires. Nominal values for the heights and span of the overhead wires were provided by AltaLink.

during the spring and summer of 2021. Hall effect sensors measure the quasi-DC component of the current flowing through the neutral connection of the transformer into and out of the ground. This measurement is used to validate our inferred GIC derived from DMM in this study.

In this paper, we introduce a new variation on the traditional DMM approach. In our deployment configuration, the remote magnetometer is instead placed close to the power line, such that the magnetic perturbation due to the GIC, $\delta\mathbf{B}(t)$, is resolvable at both the underline and remote sensors albeit with different directions and magnitudes. In the following sections, we describe an alternative DMM deployment configuration which uniquely provides two independent estimates of the GIC flowing in a HV transmission line. The derivation of these two DMM inferred GIC values is shown, and the design and deployment strategy is presented. Then, we present the first results from an application of this augmented DMM method to infer GIC on the electrical transmission network in Alberta, Canada, during a moderate geomagnetic storm on 2021-10-12. Finally, we compare the background magnetic field inferred from the DMM sensors, separated from the magnetic field associated with the GIC, to a nearby magnetometer station operated as part of the CARISMA magnetometer network. This successfully validates the augmented DMM approach.

2. An Augmented DMM Approach

The setup for the augmented DMM approach proposed and applied here is shown in the schematic diagram in Figure 1. The schematic shows the relative distances of the sensors and overhead power line wires and demonstrates the directions of the magnetic components produced by an assumed straight line current of the GIC measured at each sensor. The magnetic coordinate system used is as follows: the x component points into the page (geographic North) and parallel to the local direction of the power line, the y component points to the right (geographic East), and the z component points downward. On the right of this image, a tower carrying six overhead wires is shown.

In what follows, we make the assumption that the GIC is the same in each of the six power lines and equal to 1/6 of total GIC flowing through this power line segment. Thus, the total GIC can be calculated as a single equivalent line current flowing through a single straight wire conductor in the middle of the six wires on the tower. The GIC magnitude is assumed to be the sum of the quasi-direct current flowing through each wire located at an average height of 33 m and directly overhead of the underline sensor. Therefore, the underline magnetometer is assumed to record all of the GIC produced magnetic field in the y component, that is, $\delta B(u) = \delta B_y(u)$, and the remote sensor records the GIC produced magnetic field in both the y and z components. The same approach can be taken with a superposition of all six current elements in their actual geometric locations (see Supporting Information S1), however this introduces algebraic complexity. To aid the understanding of the reader, we restrict our analysis to a single equivalent line current (Figure 1). This augmented DMM deployment configuration offers the benefit of enabling the calculation of two independent inferred GIC values: one derived independently from each perpendicular magnetic component as a result of the current flowing through the line, $I(\Delta B_y)$ and $I(\Delta B_z)$. An example of such calculation is presented below.

2.1. Deriving an Inferred Line Element GIC

In this section, we present an approach to derive the GIC in a power line segment using this augmented DMM method. As mentioned above, for algebraic simplicity, but without loss of generality, we present a derivation for a single equivalent total GIC current, $I(t)$. As shown in Figure 1, at the location of the DMM, the overhead line comprised a total of six current elements. In the algebra presented below, these currents are all assumed to flow in a single line directly overhead at the center of the pylon in the middle of the center arm. Identical but more complicated algebra can be used to superpose the magnetic effects of all six line current elements, taking into account their actual relative locations, using the same approach presented below. To avoid unnecessarily complex trigonometric and geometric relations, we restrict ourselves to a single equivalent line current. From Ampère's law, the GIC does not produce a component along the direction of the power line in the x direction.

As seen in Figure 1, we neglect any magnetic field contribution from induced currents in the conducting Earth due to the overhead GIC. The reason for this is from the complex-image method described in Hermance and Peltier (1970) and Boteler and Pirjola (2002), a 50 A GIC is represented as a line current at a height, h, 33 m above the surface with an image current at a complex depth, $h + 2p$, where p is the complex skin depth. For a typical sedimentary basin rock represented as a halfspace with a conductivity of 10 Ωm, the skin depth at periods of 30 min and 1 min are approximately 34 and 6 km, respectively. For an image current with a period of 30 min, the magnetic contribution is approximately 0.02% and 0.2% at the underline and remote sensors, respectively. For a period of 1 min, the magnetic field contribution is 0.1% and 0.8% at the underline and remote sensors, respectively. Consistent with the approach of Matandirotya et al. (2016), we see that $h \ll p$ and the magnetic field contribution from the image current is very small; therefore, we neglect the effect of the ground conductivity in our interpretation of the measured fields at the DMM sensors.

The other unknown values are the background GMD magnetic field in each of the y and z directions, $B_{0,y}(t)$ and $B_{0,z}(t)$, which are themselves functions of time. Through a system of equations, we can solve for each and thereby separate $\mathbf{B}_0(t)$ from $\delta\mathbf{B}(t)$, where $\delta\mathbf{B}(t)$ is the magnetic field produced by the GIC and is described by the Biot-Savart Law at some perpendicular distance R from the assumed straight line GIC, $I(t)$, $\delta\mathbf{B}(t) = \frac{\mu_0 I(t)}{2\pi R}\hat{\theta}$ and where θ is the cylindrical polar coordinate around the GIC, and where $\delta\mathbf{B}(t)$ is then resolved into $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ directions according to the geometry in Figure 1. We identify the sources of the magnetic field measured at each sensor in both y and z components and define the measured fields by the following equations. The subscripts denote the magnetic field source and magnetic components in the y and z directions. The measured magnetic fields are functions of DMM sensor location, underline (u) or remote (r), and time, as seen in Figure 1.

$$B_y(u,t) = B_{0,y}(u,t) + \delta B_y(u,t) \tag{1}$$

$$B_z(u,t) = B_{0,z}(u,t) \tag{2}$$

$$B_y(r,t) = B_{0,y}(r,t) + \delta B_y(r,t) \tag{3}$$

$$B_z(r,t) = B_{0,z}(r,t) + \delta B_z(r,t) \tag{4}$$

Taking the difference between the components of the two sensors and solving for the GIC allows two independent estimates to be derived for the GIC, $I(t)$, using only the y and z components of the magnetic field, B_y and B_z, respectively, and which we denote as $I(\Delta B_y)$ and $I(\Delta B_z)$. Here, H is the vertical height of the equivalent GIC above the ground, D is the distance between the magnetometers, α is the angle from the z-axis at remote sensor to the equivalent line current, and ΔB_y and ΔB_z are the difference in the magnetic fields measured at the two sensors in the y and z components, respectively.

$$I(\Delta B_y) = \frac{2\pi}{\mu_0}\left(\frac{1}{H} - \frac{\cos(\alpha)}{\sqrt{H^2 + D^2}}\right)^{-1} \Delta B_y(t) \tag{5}$$

$$I(\Delta B_z) = \frac{2\pi}{\mu_0}\left(0 - \frac{\sin(\alpha)}{\sqrt{H^2 + D^2}}\right)^{-1} \Delta B_z(t) \tag{6}$$

We can also derive an estimate for the driving background GMD fields, $\mathbf{B}_0(t)$ using this approach. By subtracting the magnetic field of the inferred GIC from the total measured magnetic field we obtain.

$$B_{0,y}(t) = B_y(u,t) - \frac{\mu_0 I(\Delta B_y)}{2\pi}\frac{1}{H} \tag{7}$$

$$B_{0,z}(t) = B_z(r,t) - \frac{\mu_0 I(\Delta B_z)}{2\pi}\frac{\sin(\alpha)}{\sqrt{H^2 + D^2}} \tag{8}$$

The two independent DMM measurements of the inferred GIC, Equations 5 and 6, will be validated against existing Earth GIC measurements using a Hall effect sensor at the TNG connection at a proximal substation in the power grid as part of a field test of this augmented DMM technique presented in Section 4. As a secondary validation, the background GMD magnetic field, Equations 7 and 8, will be also compared against a background magnetic field measurement from the Ministik Lake (MSTK) CARISMA station, which at a distance of about 48.5 km away can be assumed to serve as a reference station for the experiment.

3. Augmented DMM System Design and Deployment

To field test the augmented DMM approach described above, we designed and implemented a test system which was capable of being deployed under power lines, and which was operated and collected data autonomously. Two Narod S100 type magnetometers mounted in waterproof Pelican cases with GPS timing synchronization comprised the DMM system, connected to a battery bank for power and a local data collection system. Three data channels, one for each component of the magnetic field, were recorded with a resolution of 25 pT at a sampling rate of 8 samples per second. This data was down-sampled to 1 sample per second, as the frequency ranges of interest for GIC are lower than 1 Hz (Boteler et al., 2019). For the first deployment from 2021 to 07–28 to 2021-10-18, two 12 V batteries were used which powered the system continuously from a fully charged state for ∼10–12 days intervals. Frequent site visits to replace the batteries and retrieve data were required during this interval. To reduce site maintenance visits, the site was upgraded with the addition of a solar panel, larger battery bank and cellular modem. This provided autonomous operation and return of data.

The batteries, computer, data logger and GPS antenna were stored in a waterproof case, top left panel in Figure 2, which was placed between the two DMM sensors. Power and serial cables ran from this central box in opposite directions to each magnetometer electronics box and sensor head. The magnetometers were leveled and orientated to point toward the local geomagnetic north and zeroed in the y component. Each sensor head was encased in a box and secured with foam to reduce any possibility of shifting orientation. No thermal insulation was used. The nominal temperature sensitivity of the S100 type magnetometers used is < 0.1 nT/°C. The sensors were mounted on heavy paving slabs placed into shallow pits dug into the soil to prevent drift over the measurement periods of interest and to protect from wind and other environmental elements, shown in the bottom left and right panels of Figure 2. The system was built to prioritize mobility and ease of deployment, and to protect the components from water damage and weather.

The field deployment location for the field test of the augmented DMM system was chosen based on the following requirements: the DMM system must record data for a power line segment where AltaLink had a nearby simultaneous TNG current measurement, essential for the validation of the DMM inferred GIC. The system must be deployed around a HV power line, 240 or 500 kV. Referring to the Hübert et al. (2020) study as a guide for the application of the DMM method on a complex power system, we aimed to deploy at a similar double-circuit power line. The DMM system is placed at least 200 m from any road or railway to avoid magnetic noise sources and in a secure location.

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Figure 2. Images from the 2021 DMM campaign show the batteries, computer, logger, and GPS antenna stored in a waterproof container (top left) which connected to each sensor by power and serial cables. Each magnetometer was mounted to a heavy paving slab which was dug into the ground (bottom left and right).

The field deployment location was chosen along a north-south segment of a 500 kV power line north and east of Edmonton, Alberta. Access and deployment in an area of land in a gated farm was generously provided by a property owner. The locations of the DMM system, AltaLink's proximal Hall effect sensors, and the permanent CARISMA MSTK magnetometer station are shown relative to the chosen segment of the 500 kV double-circuit power transmission line between the Heartland and Ellerslie substations (Figure 3). A data acquisition error of the Hall probe at Heartland transformer substation, approximately 16 km from the DMM site, required us to instead use data from the Ellerslie substation (53.434 N, 113.460 W), approximately 30 km away from our DMM location, for this study. The distance between the DMM field site and the MSTK station is 48.5 km.

The two magnetometer sensor axes are co-aligned by leveling and zeroing the magnetic east component, 14.08° from geographic north (NOAA, 2024), which we estimate was achieved with 5 nT uncertainty. The DC offset is removed and a high pass filter with a cutoff period of 4 hr is applied to remove long period daily variations in the background magnetic field not associated with GIC. The optimal position of the remote magnetometer was determined by solving for the distance between the two DMM sensors, D, at which the remote sensor would record 1 nT in the y component for a GIC of 1 A at a height of approximately 30 m. This was calculated to be approximately 71 m. For comparison, for a 1 A GIC the underline sensor would record a magnetic field perturbation of approximately 6 nT. During deployment the exact distances between the two sensors were measured to be 73.5 ± 0.5 m, varying from the optimal distance due to natural obstacles on the field site. A total positional uncertainty is derived from the errors in estimating the distance from the magnetometer to the transmission wires for each sensor. The primary assumption of the DMM technique presented here is that the power line through which the GIC is flowing is an infinitely long straight conductor at the mid-span height directly above the underline sensor. This assumption is not perfect due to the sag present in the line. From power line tower information provided by AltaLink, the assumed height of the single equivalent GIC is taken to be the average height of a weighted line that is at mid-span of the middle row of wires which is 33 ± 2 m for the segment of the electricity transmission system we deployed under. Furthermore, the sag height fluctuates with the environmental conditions and load related heating losses. By the equation for linear thermal expansion, a steel wire with the coefficient of thermal expansion of $1.2 \times 10^{-5}{}^{\circ}\mathrm{C}^{-1}$ and a variation in temperature of 20°C, the nominally 330 m long wire would expand 7.9 cm. Therefore, we assume this has a negligible effect when estimating changes to the

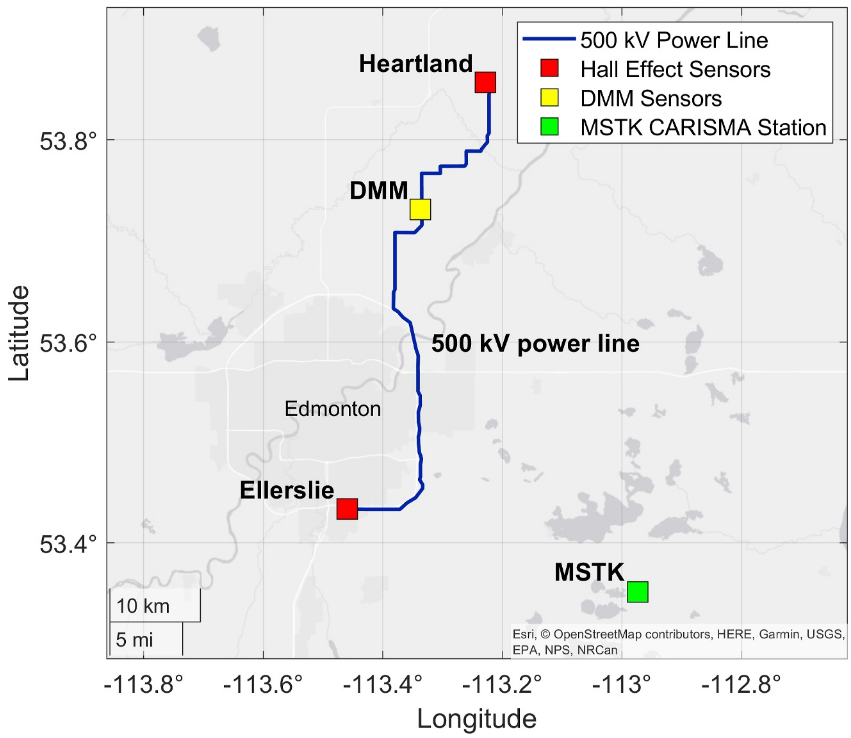

Figure 3. A map of Edmonton and surrounding areas: double circuit 500 kV power line (blue), TNG current measurement at Heartland and Ellerslie substations (red), the deployed DMM system (yellow), and the MSTK CARISMA magnetometer station (green).

sag of the line to the nearest meter. As discussed also by Hübert et al. (2020), this can be addressed using more complex algebra associated with the actual tower geometry. This is explored in more detail in the Supporting Information S1 for this paper.

The primary uncertainties in the DMM comes from the noise of the system defined by the RMS environmental noise of the magnetic field measured by the DMM sensors during a magnetically quiet period, ∼2 nT, the misalignment between the underline and remote sensor, estimated to be ∼5 nT, and the ranging error in determining the distance between each DMM magnetometer and the line current, which for the underline sensor is 2 m and for the remote sensor is 0.9 m. The noise of the fluxgate magnetometers themselves are three orders of magnitudes less than the RMS environmental noise and are therefore considered negligible. The fractional error of the distance between the GIC and the DMM sensor, R, and the magnetic field associated with the GIC, $\delta\mathbf{B}$, give us an estimate for the fractional error for I_{GIC}: $\frac{\partial I_{GIC}}{I_{GIC}} = \sqrt{\left(\frac{\partial R}{R}\right)^2 + \left(\frac{\partial(\delta B)}{\delta B}\right)^2}$. Here, $\delta\mathbf{B}$ is dependent upon the magnitude of the current, therefore the fractional error will decrease as the GIC magnitude increases. As an example, we consider a 20 A GIC which generates a magnetic field of 121 nT at the underline sensor and 51 nT at the remote sensor. An estimation for the error in $I(\Delta B_y)$ is the sum of squares of the fractional error for I_{GIC} at each DMM sensor. This is approximately 13% or 2.4 A. Since the single equivalent line current assumes $\delta B_{u,z} = 0$, an estimate for the error in $I(\Delta B_z)$ is taken to be the fractional error of I_{GIC} from only the observation of the remote sensor, approximately 11%, or 2.2 A. For a GIC of 50 A, these fractional errors would reduce to 7% and 4%, respectively. These uncertainties would be easily improved with more rigorous positional determination of the overhead wires, as it is the positional uncertainty which dominates through the propagation of errors.

4. Validation of Augmented DMM Approach

4.1. Geomagnetically Quiet Day: 2021-09-26

The magnetic measurements, in geographic coordinates and with DC offset removed, on a magnetically quiet day, 2021-09-26, at the underline, remote and MSTK sensors are plotted in the top three left panels in Figure 4. We see

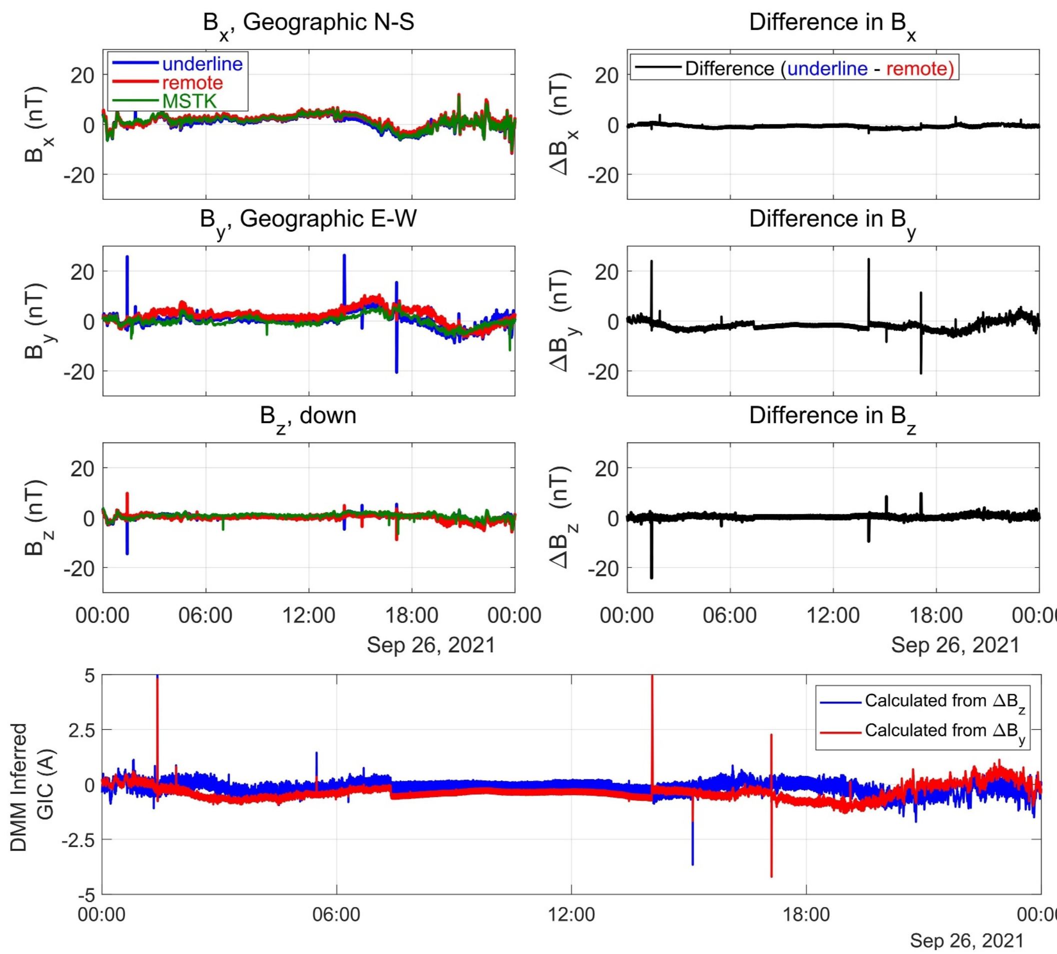

Figure 4. The magnetic field in measured by the underline and remote DMM sensors in each component (left panels), the difference between the magnetic field from the two DMM sensors in each component (right panels) and two independent measurements of the inferred GIC calculated from each of ΔB_y and ΔB_z (bottom panel) on 2021-09-26 UT.

strong agreement between the data from both DMM sensors and that from the MSTK station. This is expected for times with weak geomagnetic activity as the daily variations at two locations within a distance less than the ionospheric E-layer height of around 110 km, where the ionospheric currents are assumed to flow, should see a similar magnetic field resulting from Biot-Savart integration and cancellation magnetic effects of the smaller scale ionospheric currents overhead (Hughes & Southwood, 1976). Since the GMD is also weak, any local GIC should also be small, such that all three magnetometers might be expected to record the same magnetic field.

The comparison between these three magnetometer data sets allows an assessment of the cross calibration and relative quality and stability of the data especially arising from the temporary DMM deployment. Noticeably in the y component, the data from the two DMM sensors vary more throughout the day as compared to the MSTK sensor which remains more constant. The temporary design of the DMM deployment limits the stability of the structure supporting the DMM magnetometers and which can result in changes in sensor orientation as a result of changes in environmental conditions, whereas the MSTK magnetometer is mounted on a stable concrete block intended for permanent deployment. The difference between each of the magnetic components measured by the two DMM sensors is plotted in the top three right panels of Figure 4, revealing interesting daily variations between the data from the two DMM magnetometers. While the data from the x and z components are quite consistent across both DMM sensors, the y component sees the largest variation through the day with a maximum difference of up to 10 nT near the end of the day. We assert the cause of this 10 nT variation is most likely dominated by a misalignment of the sensor axes during deployment possibly exacerbated by diurnal misalignment due to environmental conditions, such as frost heave etc. This is observed to remain constant for different GIC

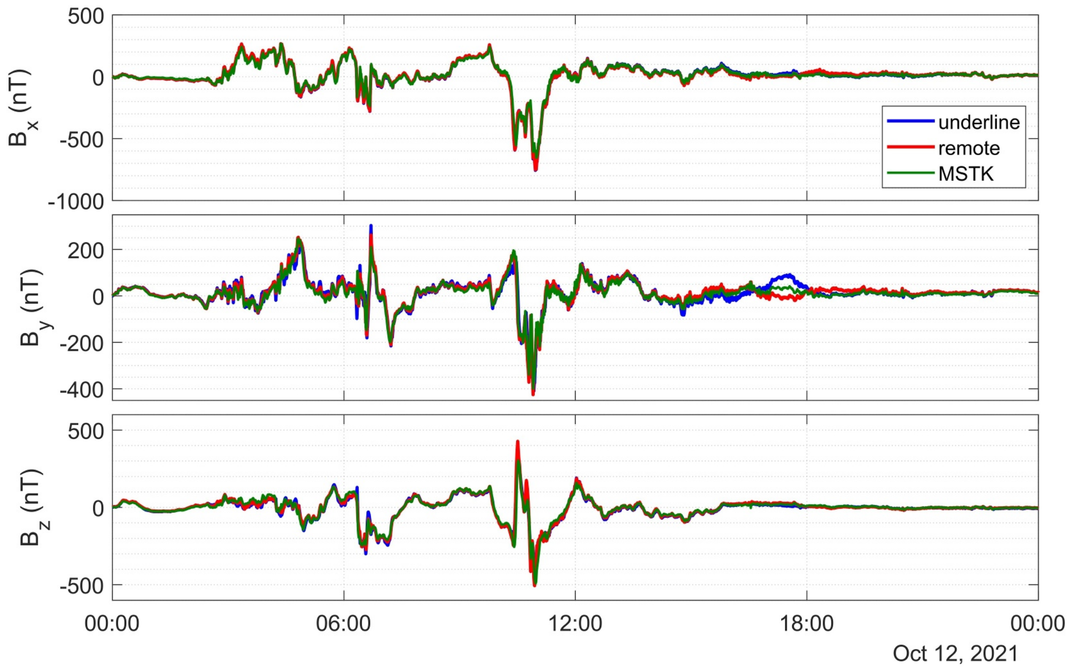

Figure 5. Magnetic field measured by the underline and remote DMM sensors and the MSTK station on 12 October 2021, a geomagnetically active day, in each component where B_x, B_y, and B_z, and where +x is North and +y is East in geographic coordinates, and z is downward.

magnitudes; therefore, this variation can be reasonably attributed to such effects and is discussed further in the following sections.

Spikes in the quiet time data were identified to be likely due to noise from either nearby vehicles or tractors. Each spike is approximately 5–10 s long and has a bipolar signal consistent with a large metal object passing by on this timescale. These signals are also only present during local daytime hours, as local sunset and sunrise occurred at 01:22 UT and 13:28 UT on this day, respectively. Overall, however, especially between around 06:00 - 15:00 UT, there is excellent agreement between all three components of magnetometer data to significantly less than 10 nT.

Using the approach from Section 3, we then use both ΔB_y and ΔB_z, second and third panels on the right side of Figure 4, respectively, to independently infer GIC in the line, bottom panel of Figure 4. During this quiet period the DMM inferred GIC is very small, <2 A. We define this as the noise floor of our DMM method, indicating that we should be able to detect GIC above this threshold; however, the error due to the sensor orientation still dominates. Nonetheless this further confirms, as expected, that the augmented DMM technique applied during times of minimal geomagnetic field disturbances infers that there is minimal GIC, when estimated using both $I(\Delta B_y)$ and $I(\Delta B_z)$.

4.2. Geomagnetically Active Day: 2021-10-12

GICs were observed in the HV electrical power network in Alberta during a moderate geomagnetic storm on 2021-10-12. The magnetic storm was triggered by a coronal mass ejection from the Sun two days prior on 2021-10-10. Upstream solar wind conditions from the DSCOVR satellite at the L1 point from the Space Weather Prediction Center showed a prolonged period of southward IMF, B_z, including a sharp increase in the magnitude of southward field coinciding with a 120 km/s uptick in the solar wind speed at 01:45 UT. The total interplanetary magnetic field magnitude reached 17 nT and a Kp of 6 was reached. The storm had a double peak in activity at approximately 06:00 UT and 12:00 UT. The geomagnetic activity on this day led to GICs in the power line providing an excellent opportunity to evaluate the effectiveness of our augmented DMM method and to validate this technique against Hall effect sensor GIC data from TNG connections in the AltaLink network. Figure 5 shows the time series of the x, y, and z components of the magnetic field, in geographic coordinates, as measured by the underline and remote sensors of the DMM system, and by the magnetometer at MSTK station on 2021-10-12. The

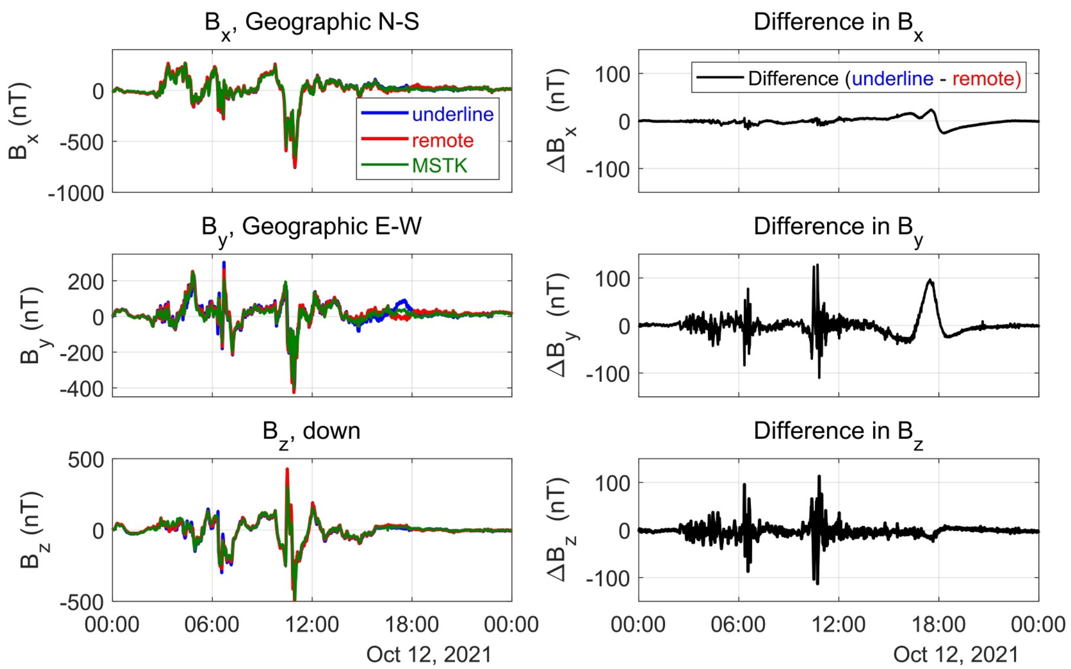

Figure 6. Magnetic field in geographic coordinates measured by the underline and remote DMM sensors in each component (left) and the difference between the sensors in each component (right) on 12 October 2021.

notable difference as compared to Figure 4 is the magnitudes of the GMD as seen by all three magnetometers. MSTK provides a strong proxy for the driving GMD, but its values deviate from the differential magnetometer measurements, particularly in the y component around the times of the strongest magnetic disturbance, as would be expected if the DMM stations are additionally measuring magnetic perturbation associated with local and overhead GIC in the power line. There also appears to be an interesting feature at around 17:00 to 19:00 UT where all three stations have significantly different magnetic fields. There is also some small baseline divergence between the two DMM magnetic fields, but overall the data appears amenable for DMM analysis.

4.3. Measuring GICs Using DMM

The magnetic field in each component at the underline and remote DMM magnetometers, and the component-wise difference, is shown in Figure 6 in the left and right columns, respectively. The peak-to-peak amplitude of the difference in the x component, the component parallel to the power line and the direction of the GIC flow, and for which we assume a perfectly straight infinite line current, is less than 20 nT, disregarding the anomaly at 14:00 to 19:00 UT which we attribute to temperature-related environmental sensor-alignment effects. The parallel field, ΔB_x, as expected, is close to zero as a consequence of the location of the sensor and the assumed geometry of the GIC flowing locally north-south in the power line. This in itself is significant as it confirms that the difference in the magnetic field recorded by each DMM sensor in the y and z components can confidently be predominantly associated with the GIC flowing in overhead transmission lines. This further provides an initial estimate for the errors in $\Delta B_{y,z}$. We consider that the predominant source of error in the DMM will be due to the pointing uncertainty of the sensors, as previously examined in Sections 3 and 4.1.

An anomalous variation in the magnetic field measured by the DMM sensors occurs during the late hours of the day. While the total magnetic field remains constant, after removing the DC offset and applying a high pass filter, this variation is most visible in the y component around 14:00 to 19:00 UT. On 2021-10-12, the minimum temperature reached was −5.9°C at 14:00 UT and the maximum temperature was 9.8°C at 21:00 UT. Below freezing temperatures were maintained from 01:00 to 13:00 UT, according to historical climate data from Environment and Natural Resources Canada (https://climate.weather.gc.ca). The anomaly is coincident with the temperature reaching the dew point in the early morning. We conclude that this anomaly is due to the freezing and thawing of condensation beneath the DMM sensor and the concrete slab on which it is placed. This could cause a rotation or tilt leading to misalignment between the orientation of the two DMM magnetometers and changing the

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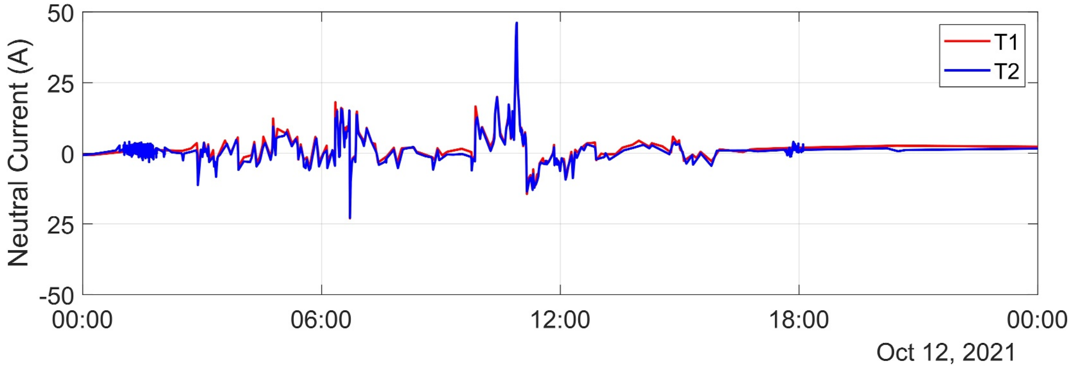

Figure 7. The TNG current measured on each circuit, T1 and T2, at Ellerslie substation on 2021-10-12.

amplitude of the magnetic field measured in each component. For a change in B_y of approximately 110 nT, the rotation in the x and y plane is around 0.5°. Fortunately, the rotation caused by condensing morning dew occurred outside the period of the strongest geomagnetic activity, providing stable magnetic field data for this time interval. During the period of stability, as seen in Figure 6, the largest peak-to-peak DMM amplitudes of ΔB_y was 136 and 238 nT at around 06:40 UT and 10:45 UT, respectively. Similar peak-to-peak DMM amplitudes are seen in ΔB_z: 152 nT at 06:40 UT and 226 nT at 10:45 UT.

We now compare the DMM inferred GIC to the GIC measured by the Hall probe at the TNG connection. The two (sets of 3 single-phase) transformers at the end of each circuit on the 500 kV power line each have a ground connection and are referred to as T1 and T2. The TNG currents measured at both ground connection points have similar amplitudes, as seen in Figure 7; therefore, we assume that equal GIC flows in both circuits. A simplified circuit diagram, Figure 8, shows the relative position of the DMM system along the power line segment, the Hall effect sensors on T1 and T2 at Ellerslie substation, and the power lines connecting into Ellerslie substation. The comparison between the DMM GIC and Hall sensor GIC inherently assumes that all the GIC in the line flows to ground through T1 and T2, but this is not true as this is not an isolated system and GIC may flow through other connections in the HV grid or to the lower voltage distribution portion of the network. The total GIC at Ellerslie is calculated by adding the TNG current measured by the Hall sensors on both T1 and T2.

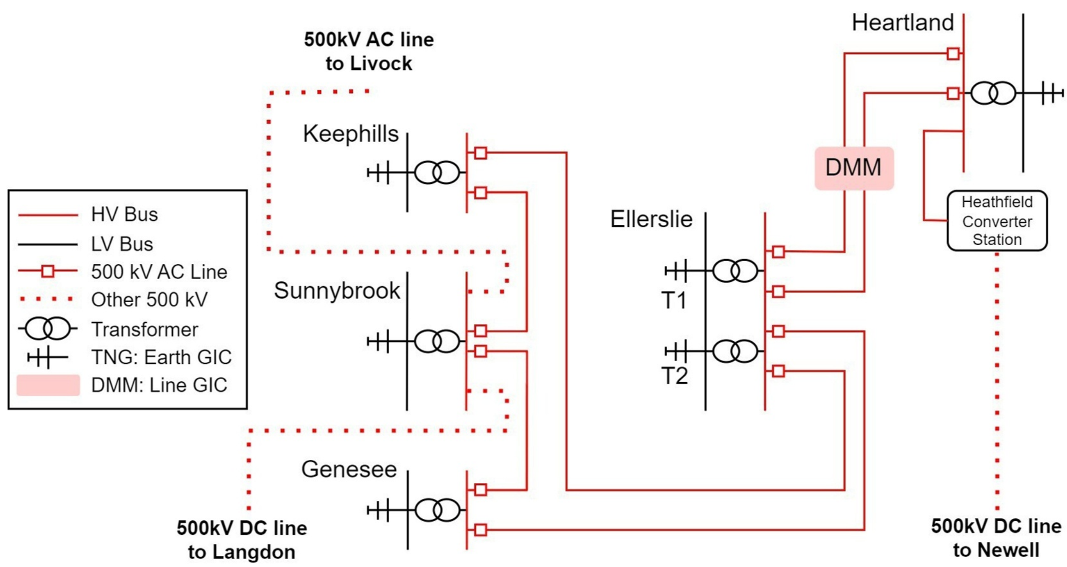

Figure 8. Simplified circuit diagram of a portion of the 500 kV network around Edmonton, Alberta shows the location of the DMM system and the TNG measurements (T1 and T2) at Ellerslie substation. The dashed red line shows 500 kV transmission lines that connect the Edmonton region to elsewhere in Alberta.

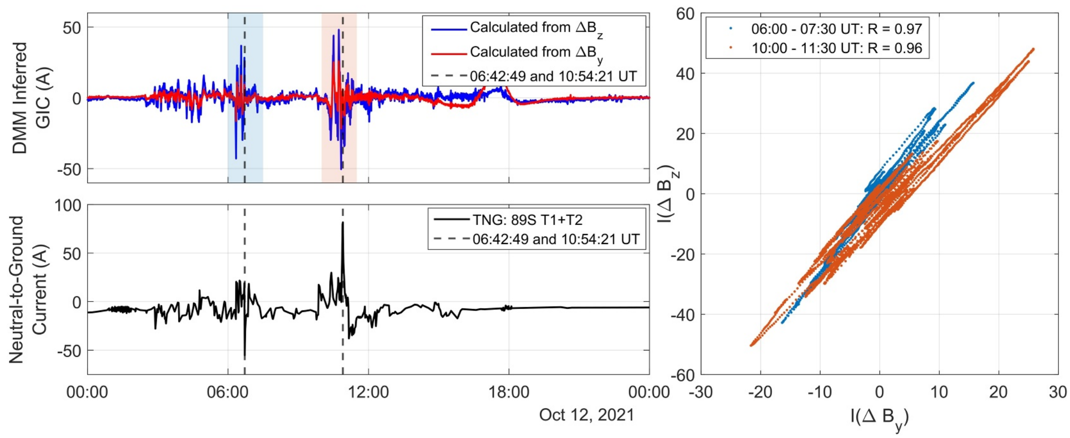

Figure 9. (Top left) Two independent measurements for the total in-line DMM-inferred GIC calculated from the ΔB_z and ΔB_y components are plotted, blue and red traces, respectively. (Bottom left) The total TNG current measured at Ellerslie substation (T1 + T2—see text for description). (Right) $I(\Delta B_y)$ is plotted against $I(\Delta B_z)$ during two strongest GMD active times, 06:00–07:30 UT and 10:00–11:30 UT (shaded in top left panel), with a linear correlation between the two DMM-inferred GIC values during these times.

The two independent estimates of the inferred GIC, $I(\Delta B_y)$ and $I(\Delta B_z)$, were computed using the steps outlined in Section 2. The inferred GIC is calculated independently from the y and z components, each providing an estimate of the GIC associated with each component of the magnetic field perpendicular to the direction of the current using Equations 5 and 6. In Figure 9, $I(\Delta B_y)$ and $I(\Delta B_z)$ in Amperes (A) (top panel) are compared to the total TNG current measured at Ellerslie transformer substation by AltaLink (bottom panel; also in Amperes).

First, in the top panel of Figure 9, we see that the two DMM-inferred GIC from each perpendicular component, $I(\Delta B_y)$ and $I(\Delta B_z)$, have excellent temporal and waveform correspondence. This is especially true at the times of strongest geomagnetic activity, between around 06:00 to 07:30 UT and from 10:00 UT to 11:30 UT. Between these hours, the linear correlation coefficient between the two time series, $I(\Delta B_y)$ and $I(\Delta B_z)$, is 0.97 and 0.96, respectively. The DMM-inferred GIC calculated from the difference in the y component has a smaller peak amplitude than that calculated from the difference in the z component. This difference can be partly attributed to a combination of the error in the power line height assumption and the near field cancellation effect, both of which have a larger effect on ΔB_y (discussed further in Text S1 in Supporting Information S1). Since the GIC is estimated using the assumption of an infinitely long single straight equivalent current directly above the underline sensor, we assume that the magnetic signature of the GIC in the z component at the underline sensor is negligible and all of $\delta\mathbf{B}(t)$ should be in the y component. By this assumption, Equation 6 is determined independently from $\delta\mathbf{B}$ at the underline sensor, thus effectively treating the underline sensor as a base station for inferred GIC in the z component, ΔB_z. This also introduces a small error to the magnitude in each of $I(\Delta B_y)$ and $I(\Delta B_z)$ due to misalignment in the y-z plane that is assumed to be small. Further, the difference in GIC magnitude calculated with the single equivalent line assumption (Equations 5 and 6) and with all the line current elements resolved (see Text S1) is purely geometric and therefore a scaling factor to convert from one to the other can be calculated. The single equivalent line current assumption overestimates $I(\Delta B_y)$ by a factor of 1.089, and underestimates the $I(\Delta B_z)$ by a factor of 0.996 for the power line geometry at the DMM deployment location. At the time of peak disturbance amplitude, this factor accounts for a difference of 2.2 A in $I(\Delta B_y)$ and 0.15 A in $I(\Delta B_z)$. Details about the derivation of the equations for the individual current elements of the GIC and the calculation of the scaling factor can be found in the Supporting Information S1 for this paper.

In comparing $I(\Delta B_y)$, $I(\Delta B_z)$ and the TNG current measurement, it is clear the largest peaks in the Hall probe data are contemporaneous with peaks in the DMM GIC data. Clear peaks in the amplitude of the DMM-inferred GIC are seen occurring at the times of the maximum TNG current observed at 06:42 UT and 10:54 UT, as indicated by the black dashed vertical lines in Figure 9. It is important to note that the TNG current data at Ellerslie substation has a nonuniform sample rate resulting in data gaps which are then linearly interpolated and thus fails to show some higher frequency information of low amplitude GIC. Despite the TNG current data being discontinuous, the waveform and temporal correspondence of the GIC in the line inferred from the DMM method and the TNG

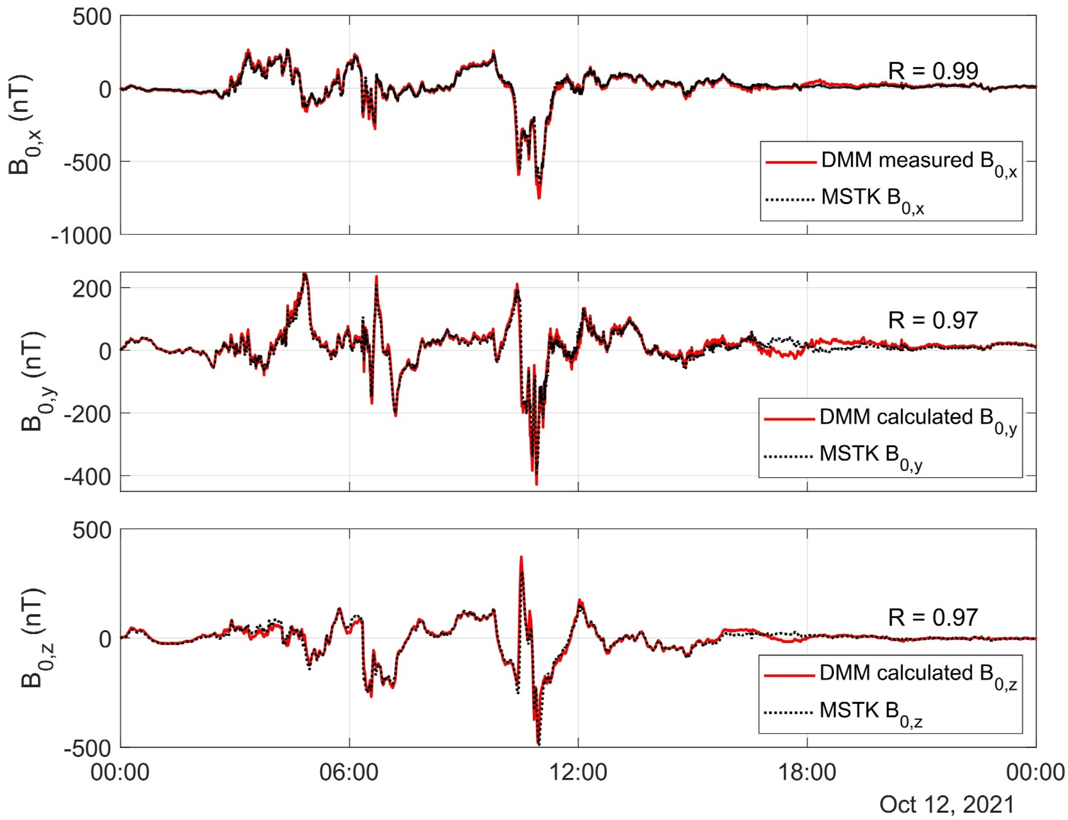

Figure 10. The background magnetic field determined using the DMM method is plotted in the x, y, and z components. MSTK station magnetic field is also plotted in each component for comparison.

current measured by the Hall probe are in good agreement. Further, their amplitudes are of the same magnitude. Note that, an estimate of the GIC flowing per phase is calculated by dividing the GIC by six. The maximum total GIC amplitude estimated by the DMM method was 54 A, or 9 A per phase, while the maximum GIC measured by the Hall probes at the TNG connection was 86 A or about 14 A per phase.

In making this direct comparison we are also assuming that, at the termination of the line at Ellerslie substation, all the GIC is returned to ground. The discrepancy between the DMM-inferred GIC measurement and that from the TNG can hence likely only be partly attributed to the positional uncertainties and sensor misalignment. Elements which can be used to determine the GIC flowing into Ellerslie from other grid connections, but for which we lack the required information and therefore are not considered in this study, include the grid topology and the resistances of the transformers and power lines (cf. the discussion in Hübert et al., 2020). These elements may play an important role in comparing the amplitude of the GIC flowing through the TNG connection versus the power line.

4.4. Reconstruction of GMD Using DMM Data

As a final validation of the augmented DMM technique, we reconstruct the background GMD using the derived Equations 7 and 8. In each of the y and z magnetic field components, the magnetic field produced by the GIC is identified and the background GMD is then determined by subtracting the field, inferred to be related to the GIC by DMM in the power line, from the total field measured. In Figure 10, the reconstructed GMD is compared to the magnetic field measured at the MSTK CARISMA magnetometer station. Figure 10 also shows excellent linear correlation between the magnetic field measured at MSTK and each $B_{0,x}$, $B_{0,y}$, and $B_{0,z}$ reconstructed using the differential measurements. For each magnetic component, the linear correlation coefficient is determined to be $R = 0.99$, $R = 0.97$ and $R = 0.97$, respectively. This is strong evidence for the efficacy of this augmented DMM technique in separating the GMD field from the magnetic perturbations associated with GIC.

Two important points should be considered when accounting for the difference between these two measurements. Firstly, this comparison holds the assumption that the GMD is spatially uniform over a distance of approximately 48 km, and secondly, there are discrepancies associated with base lining which can result from changes in the orientation of the DMM sensors, such as that caused by environmental factors. There also appears to be magnetic signatures at MSTK which are not captured by the DMM and/or vice versa. Therefore, even though we have demonstrated MSTK is a good proxy for the GMD at the DMM site, as we have shown, the small differences between two magnetometer measurements can of course affect the magnitude of GIC values estimated by the DMM method. Likely, MSTK station is too far away from the DMM site to be used as a true reference station.

5. Discussion and Conclusions

Using the known geometry of this modified DMM system, we have demonstrated how we were able to separate the magnetic field associated with the GIC flowing through the power line and the temporal variations in the background GMD, despite both signals being present in both DMM magnetometer data sets. An advantage of this modified method is that we can compute two independent estimates for the inferred GIC by using the difference in each of the magnetic y and z components separately. Therefore, as we have shown here, through a comparison of the two independent estimates for the DMM-inferred GIC, we confirmed the amplitude difference between the two GIC estimates could be attributed to the sources of error presented in Section 3, particularly the positional errors from assuming nominal line heights, unequal GIC in the six overhead lines, or displacement of the underline sensor from the center of the tower. Using data from a magnetically quiet period we were also able to demonstrate good intercalibration between the measurement from the two DMM sensors. We further validated the temporal correspondence and magnitude of the two GIC estimates from DMM against the TNG current measured by AltaLink at a proximal substation in the Alberta power grid.

Using TNG current data collected by a Hall effect sensor deployed by AltaLink at the Ellerslie transformer substation, we validated the augmented DMM method by comparing the measured GIC from the Hall probe with our GIC derived using DMM. We see that the largest peaks in the TNG current data at 06:42:49 UT and 10:54:21 UT are contemporaneous with large peaks in the DMM GIC data set. The total peak GIC measured using the Hall probe is 86 A, or 14 A per phase. The maximum GIC values flowing in the transmission line estimated using the DMM approach was 54 A or 9 A per phase. Although this is a significant GIC considering this was a moderate geomagnetic storm, it is not near the threshold defined by the North American Electric Reliability Corporation to be a GIC of concern at 75 A per phase (NERC, 2020). The difference between the inferred GIC from DMM and the TNG current may be partly attributed to the misalignment and positional uncertainty introduced to the DMM system during deployment, and the unknown resistances of the transformer and the power line itself. Further, as can be seen in Figure 8, there are multiple connections at Ellerslie substation and therefore additional GIC traveling into the substation from these lines could be further attributing to the difference between the DMM GIC and TNG current. Comparison to an HV network model of the Alberta power network is needed to understand the GIC flowing to ground at Ellerslie substation (cf. for example the recent paper by Cordell et al., 2024). Missing information about the resistance parameters of the power network limits further comparison (cf. Hübert et al., 2020; Hübert et al., 2024).

Further, we compared the background GMD derived from the augmented DMM method to the magnetic field measured at the nearby MSTK CARISMA station, the two data sets showing good agreement. We see some small scale differences during large GMD which suggests large background GMD includes smaller scale currents which generates GMD with slightly varying amplitudes between MSTK station and the DMM location. The background GMD determined at both locations has correlation coefficients greater that 0.97 in each component for the entire active day. This further suggests that the equations derived in Section 2 properly separates the two magnetic field sources measured by the DMM sensors and supports our conclusions above for GIC determined using the augmented DMM method. However, although MSTK station can be used as a proxy to estimate GMD driving the GIC in the electrical power system, it is probably too far away from the DMM site to be used as a true base station or an effective third DMM sensor.

Overall, the strengths of an augmented DMM approach have been demonstrated by the following results presented in this study.

- Validated large ΔB, and GIC, come from periods of large GMD;
- Validated independent estimates of GIC can be derived from ΔB_y and ΔB_z independently;

- Demonstrated temporal correspondence between the DMM inferred GIC and the TNG current in the power grid despite the nonuniformly sampled character of the TNG data at this location;
- Validated success of the augmented DMM inferred GIC by showing the derived background GMD, B_0, compares well with the magnetic field at a nearby permanent magnetic station.

6. Future Work

In the near term, we plan to continue GIC monitoring using multiple new DMM stations with applied upgrades for more permanent deployment and continuous remote operation. This will demonstrate the usefulness of the DMM method on a wider scale for future use by industry. Further improvements to the DMM system in the future will increase the stability of the sensor deployment against environmental factors. We recommend future DMM deployment could include a third DMM sensor as a local reference station, at approximately 100–200 m away from the underline sensor, while keeping the underline and intermediate magnetometers at the locations demonstrated in our design. This would be a similar approach to Hübert et al. (2020) and Matandirotya et al. (2016), but would maintain the benefits of providing two independent GIC estimates as demonstrated here. Lastly, we plan to utilize a recently developed HV network model (Cordell et al., 2024) to make appropriate comparisons between the GIC in the power line and the GIC flowing to ground in the substation and to inform future DMM deployment locations.

Based on the results presented here, we are confident that the DMM approach can represent a valuable tool for power transmission facility operators to assess GIC flows and systematic vulnerabilities in the electric power grid. For example, a temporary DMM deployment can be used to assess potential GIC hotspots. Since DMM deployments are completely passive, this can be done without any impacts on the electric power operations. Results from such DMM deployments might validate GIC hotspots inferred from models, or validate choices for locations for more permanent deployment of Hall sensors for additional TNG GIC monitoring.

Data Availability Statement

Magnetic field data for the Ministik Lake (MSTK) magnetic station was provided by CARISMA and is freely available at CARISMA Data (2024). The presented DMM magnetometer data and Hall probe data is hosted in a repository at Parry (2024). Ellerslie and Heartland substation and transmission line locations are available at Openstreetmap (2024).

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References

Bailey, R. L., Halbedl, T. S., Schattauer, I., Achleitner, G., & Leonhardt, R. (2018). Validating GIC models with measurements in Austria: Evaluation of accuracy and sensitivity to input parameters. *Space Weather*, *16*(7), 887–902. https://doi.org/10.1029/2018sw001842

Blake, S. P., Gallagher, P. T., Campanyà, J., Hogg, C., Beggan, C. D., Thomson, A. W. P., et al. (2018). A detailed model of the Irish high voltage power network for simulating GICs. *Space Weather*, *16*(11), 1770–1783. https://doi.org/10.1029/2018sw001926

Bolduc, L. (2002). GIC observations and studies in the Hydro-Québec power system. *Journal of Atmospheric and Solar-Terrestrial Physics*, *64*(16), 1793–1802. https://doi.org/10.1016/s1364-6826(02)00128-1

Boteler, D. H. (2019). A 21st century view of the March 1989 magnetic storm. *Space Weather*, *17*(10), 1427–1441. https://doi.org/10.1029/2019sw002278

Boteler, D. H., & Pirjola, R. J. (2002). The complex-image method for calculating the magnetic and electric fields produced at the surface of the Earth by the auroral electrojet: The complex-image method. *Geophysical Journal International*, *132*(1), 31–40. https://doi.org/10.1046/j.1365-246x.1998.00388.x

Boteler, D. H., & Pirjola, R. J. (2017). Modeling geomagnetically induced currents. *Space Weather*, *15*(1), 258–276. https://doi.org/10.1002/2016sw001499

Boteler, D. H., Pirjola, R. J., & Marti, L. (2019). Analytic calculation of geoelectric fields due to geomagnetic disturbances: A test case. *IEEE Access*, *7*, 147029–147037. https://doi.org/10.1109/access.2019.2945530

Boteler, D. H., Pirjola, R. J., & Nevanlinna, H. (1998). The effects of geomagnetic disturbances on electrical systems at the Earth's surface. *Advances in Space Research*, *22*(1), 17–27. (Publisher: Elsevier BV. https://doi.org/10.1016/s0273-1177(97)01096-x

Butala, M. D., Kazerooni, M., Makela, J. J., Kamalabadi, F., Gannon, J. L., Zhu, H., & Overbye, T. J. (2017). Modeling geomagnetically induced currents from magnetometer measurements: Spatial scale assessed with reference measurements. *Space Weather*, *15*(10), 1357–1372. https://doi.org/10.1002/2017sw001602

Campbell, W. H. (1980). Observation of electric currents in the Alaska oil pipeline resulting from auroral electrojet current sources. *Geophysical Journal International*, *61*(2), 437–449. https://doi.org/10.1111/j.1365-246x.1980.tb04325.x

CARISMA Data. (2024). CARISMA data. [Dataset]. https://carisma.ca/

Cordell, D., Mann, I. R., Parry, H., Unsworth, M. J., Cui, R., Clark, C., et al. (2024). Modeling geomagnetically induced currents in the Alberta power network: Comparison and validation using hall probe measurements during a magnetic storm. *Space Weather*, *22*(4), e2023SW003813. https://doi.org/10.1029/2023sw003813

Cordell, D., Unsworth, M. J., Lee, B., Hanneson, C., Milling, D. K., & Mann, I. R. (2021). Estimating the geoelectric field and electric power transmission line voltage during a geomagnetic storm in Alberta, Canada using measured magnetotelluric impedance data: The influence of three-dimensional electrical structures in the lithosphere. *Space Weather*, *19*(10), e2021SW002803. https://doi.org/10.1029/2021sw002803

Haque, A., Vaile, J., Rutkunas, T., Kodsi, S., Bhuiya, A., & Baker, R. (2017). Geomagnetic disturbance storm system impact — A transmission facility owner case study. In *2017 IEEE power & energy society general meeting* (pp. 1–5). https://doi.org/10.1109/PESGM.2017.8274485

Hermance, J. F., & Peltier, W. R. (1970). Magnetotelluric fields of a line current. *Journal of Geophysical Research*, *75*(17), 3351–3356. https://doi.org/10.1029/jb075i017p03351

Hübert, J., Beggan, C. D., Richardson, G. S., Gomez-Perez, N., Collins, A., & Thomson, A. W. P. (2024). Validating a UK geomagnetically induced current model using differential magnetometer measurements. *Space Weather*, *22*(2), e2023SW003769. https://doi.org/10.1029/2023sw003769

Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. *Space Weather*, *18*(4), e2019SW002421. https://doi.org/10.1029/2019sw002421

Hughes, W. J., & Southwood, D. J. (1976). The screening of micropulsation signals by the atmosphere and ionosphere. *Journal of Geophysical Research*, *81*(19), 3234–3240. https://doi.org/10.1029/ja081i019p03234

MacManus, D. H., Rodger, C. J., Dalzell, M., Thomson, A. W. P., Clilverd, M. A., Petersen, T., et al. (2017). Long-term geomagnetically induced current observations in New Zealand: Earth return corrections and geomagnetic field driver. *Space Weather*, *15*(8), 1020–1038. https://doi.org/10.1002/2017sw001635

Mäkinen, J. T. T. (1993). *Geomagnetically induced currents in the Finnish power transmission system* (Vol. 32, p. 101). Finnish Meteorological Institute Geophys. Publications.

Mann, I. R., Milling, D. K., Rae, I. J., Ozeke, L. G., Kale, A., Kale, Z. C., et al. (2008). The upgraded CARISMA magnetometer array in the THEMIS era. *Space Science Reviews*, *141*(1–4), 413–451. https://doi.org/10.1007/s11214-008-9457-6

Marsal, S., Torta, J. M., Curto, J. J., Canillas-Pérez, V., Cid, O., Ibañez, M., & Marcuello, A. (2021). Validating GIC modeling in the Spanish power grid by differential magnetometry. *Space Weather*, *19*(12), e2021SW002905. https://doi.org/10.1029/2021sw002905

Marshall, R. A., Dalzell, M., Waters, C. L., Goldthorpe, P., & Smith, E. A. (2012). Geomagnetically induced currents in the New Zealand power network. *Space Weather*, *10*(S8), S08003. https://doi.org/10.1029/2012sw000806

Matandirotya, E., Cilliers, P. J., Zyl, R. R. V., Oyedokun, D. T., & Villiers, J. (2016). Differential magnetometer method applied to measurement of geomagnetically induced currents in Southern African power networks. *Space Weather*, *14*(3), 221–232. https://doi.org/10.1002/2015sw001289

Molinski, T. S. (2002). Why utilities respect geomagnetically induced currents. *Journal of Atmospheric and Solar-Terrestrial Physics*, *64*(16), 1765–1778. https://doi.org/10.1016/s1364-6826(02)00126-8

NERC. (2020). Transmission system planned performance for geomagnetic disturbance events. (Tech. Rep. No. TPL-007-4). *North American Electric Reliability Corporation*.

Nikitina, L., Trichtchenko, L., & Boteler, D. H. (2016). Assessment of extreme values in geomagnetic and geoelectric field variations for Canada. *Space Weather*, *14*(7), 481–494. https://doi.org/10.1002/2016SW001386

NOAA. (2024). NOAA mobile magnetic declination calculator [Software]. https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml

Openstreetmap. (2024). Openstreetmap [DataSet]. https://www.openstreetmap.org/

Parry, H. G. (2024). Alberta differential magnetometer measurement (DMM) data from 2021 [Dataset]. *Zenodo*. https://doi.org/10.5281/zenodo.11188122

Pirjola, R. (2000). Geomagnetically induced currents during magnetic storms. *IEEE Transactions on Plasma Science*, *28*(6), 1867–1873. https://doi.org/10.1109/27.902215

Price, P. (2002). Geomagnetically induced current effects on transformers. *IEEE Transactions on Power Delivery*, *17*(4), 1002–1008. https://doi.org/10.1109/tpwrd.2002.803710

Pulkkinen, A., Thomson, A., Clarke, E., & McKay, A. (2003). 2000 geomagnetic storm: Ionospheric drivers of large geomagnetically induced currents. *Annales Geophysicae*, *21*(3), 709–717. https://doi.org/10.5194/angeo-21-709-2003

Pulkkinen, A., Viljanen, A., Pajunpää, K., & Pirjola, R. (2001). Recordings and occurrence of geomagnetically induced currents in the Finnish natural gas pipeline network. *Journal of Applied Geophysics*, *48*(4), 219–231. https://doi.org/10.1016/s0926-9851(01)00108-2

Rosenqvist, L., & Hall, J. O. (2019). Regional 3-d modeling and verification of geomagnetically induced currents in Sweden. *Space Weather*, *17*(1), 27–36. https://doi.org/10.1029/2018sw002084

Thomson, A. W. P., McKay, A. J., Clarke, E., & Reay, S. J. (2005). Surface electric fields and geomagnetically induced currents in the Scottish power grid during the 30 October 2003 geomagnetic storm. *Space Weather*, *3*(11), S11002. https://doi.org/10.1029/2005sw000156

Torta, J. M., Marcuello, A., Campanyà, J., Marsal, S., Queralt, P., & Ledo, J. (2017). Improving the modeling of geomagnetically induced currents in Spain. *Space Weather*, *15*(5), 691–703. https://doi.org/10.1002/2017sw001628

Trivedi, N. B., Vitorello, Í., Kabata, W., Dutra, S. L. G., Padilha, A. L., Bologna, M. S., et al. (2007). Geomagnetically induced currents in an electric power transmission system at low latitudes in Brazil: A case study. *Space Weather*, *5*(4), S04004. https://doi.org/10.1029/2006sw000282

Viljanen, A., & Pirjola, R. (1994). Geomagnetically induced currents in the Finnish high-voltage power system: A geophysical review. *Surveys in Geophysics*, *15*(4), 383–408. https://doi.org/10.1007/bf00665999

Watari, S., Kunitake, M., Kitamura, K., Hori, T., Kikuchi, T., Shiokawa, K., et al. (2009). Measurements of geomagnetically induced current in a power grid in Hokkaido, Japan. *Space Weather*, *7*(3), S03002. https://doi.org/10.1029/2008sw000417